

It would require considerable time and energy to enable the mobile app to directly access the variety of data sources being used at the enterprise backend.

Enterprise data security

With legacy systems and related security ecosystems, enterprise IT heads can remain relaxed because all the enterprise's data resides in and is mostly used from within the enterprise premises and via enterprise-owned devices like laptops, desktops, etc.

With mobility, data not only crosses beyond the enterprise premises, but often also resides on the users' personal mobile devices for offline access. Further, enterprise data communication happens mostly over untrusted networks on these mobile devices, which creates the risk of enterprise data being stolen or tampered with. Also, confidential enterprise data may also get copied, printed, etc. Other malicious apps may invade the enterprise app sandbox and access its data. The user may lose the device containing enterprise data, leading to a leak of the confidential data residing in the device. Developing a robust and reliable solution to secure both enterprise data at rest and data-on-the-move pertaining to the app (while not violating the user's privacy) is very crucial.

Remote management

Any loophole in an app's life cycle and its data management can make an enterprise vulnerable to mismanagement of app versions, users and their devices. For instance, the app should support a way to enforce app data wiping if a user is leaving the company or has lost the device. Further, if the enterprise wants to make a previous app version obsolete and prohibit its use, then it should be able to enforce specific app version updates as mandatory on the target user base.

Only complete control and management of the app's life cycle, together with the installed app and its data, can ensure the solution generates the desired return on investment (RoI) for an enterprise.

User adoption

In today's mobility era, an app's success depends, to a great extent, on its user experience. Streamlined use cases, ease of use, shorter workflows, a focus on only what the user needs in the current context, and a respect for screen size contribute to a better user experience. The better the user experience, the higher the chances of more users adopting the app. And the lower the user adoption, the higher the chance of an enterprise failing in its mobile app journey.

Enterprise mobility, as a segment, has been maturing over the past few years, and now there are solution suites available to address these challenges effectively. Enterprise mobility management (EMM) and the mobile application development platform (MADP) are well known solutions in the market. In this article, we will have a look at MADP and how it solves some of these challenges.

The solution: MADP

According to Gartner, "A MADP enables an enterprise to design, develop, test, deploy, distribute, analyse and manage a portfolio of cross-platform mobile apps on a range of devices running Android and iOS, and to address the requirements of diverse use cases including external-facing and internal-facing scenarios."

There are generic MADPs available in the market as products or by service providers as their solution catalysts. Whether to use a generic MADP or opt for a custom one largely depends on the complexity of your enterprise's ecosystem, its time-to-market needs, security concerns and customisation requirements.

Figure 1 shows the market leading MADPs, as per Gartner's Magic Quadrant for MADPs, July 2015.

As per this report, the MADP market continues to mature, moving away from the tech-centric position of the past towards a more business-aligned approach, expanding support for the app development life cycle and offering more robust and scalable capabilities.

As with all Magic Quadrants, vendors in the 'Leaders quadrant' are not necessarily the best for all projects or enterprises. Depending on an enterprise's needs, a vendor in any quadrant could be the best for an enterprise. Some platforms may be a better fit with an organisation's particular skill sets. Consider the match between the skills available and those required by each MADP evaluated. Also consider the cost and complexity of the vendor's offering, and whether this aligns with benefits expected during the estimated life span of the product.

Some enterprises may prefer custom MADP, to address specific concerns related to vendor lock-in, the nature of the solution, security, roadmap demands, long-term costs, etc. For developing custom MADP, enterprises can consider



Figure 1: Gartner Magic Quadrant for MADPs in 2015